

Linear Algebra

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1 Preliminary

The classical linear algebra relies heavily on 'basis' and studies how a matrix could be pretty. This reliance on basis brings heavy calculations when switching a basis to another and covers the essence of the operator. Therefore, we pursue a basis-free language to redo the linear algebra. This reformulation will make the theory of linear algebra concise and compact.

2 The structure of operators

Let V be a finite dimensional vector space over a field k .

In classical linear algebra, we would ask what kind of operators are diagonalizable. The theories are heavy there. From those theories, we notice that it has some characterization of decomposing the space V associated with linear map $T : V \rightarrow V$ into direct sums of one dimensional subspaces.

So, we can thought of T as action on V . This action can be extended linearly to an action of $k[T]$ on V . Together with the abelian group structure of V , V has a $k[T]$ -module structure.

Now, we have two perspectives: One is the ring $k[T]$. This involves the characteristic polynomials theories in classical linear algebra theory. Another is the $k[T]$ -module V . This entails the decomposition of V . In this manner, the heavy and redundant classical theories are packed up in these two perspectives.

Under the context of $k[T]$ -modules, operators (representing $k[T]$) interacts with the space V .

2.1 Characteristic polynomials: Study of $k[T]$

2.2 Decomposition of $k[T]$ -modules: for specific T

The question of asking whether an linear operator $T : V \rightarrow V$ is diagonalizable or trianguable, is asking what kind of operators T make the corresponding $k[T]$ -module V nicely decomposed.

So, we give a definition

Definition 2.1: Diagonalizable operator

An operator $T \in \text{End}_k(V)$ is **diagonalizable** $\Leftrightarrow k[T]$ -module V is semisimple.

Definition 2.2: Trianguable operator

An operator $T \in \text{End}_k(V)$ is **trianguable** $\Leftrightarrow k[T]$ -module V has a composition series.

This is one of the two perspectives mentioned. From the perspective of $k[T]$, the characteristic polynomials, we obtain a characterization of this definition

Corollary 2.1: characterization of diagonalizable operator

2.3 Decomposition of $k[T]$ -modules: for arbitrary T (over PID)

The previous questiones fix a certain decomposition of V and ask what kind of operator T has a decomposition that is compatible with the given decomposition. Now, we reverse a bit: how to decompose a space $k[T]$ -module V in general for any T ?

Before asking this question, all the theories are on vector spaces V over a field k . This ambient condition could be generalized to some extent, till a module over a PID. To avoid the redundancy, the theory will be written in the context of modules over PID and the extra properties on k -vector space will be listed additionally.

3 The geometry of spaces

3.1 Bilinear form

We start to introduce bilinear form in order to classify all kinds of 'product' geometry. A bilinear form is a map $f : V \times V \rightarrow k$ that is linear in each slot.

intermediate

3.1.1 Classification of classic bilinear forms (on $\mathbb{R}, \mathbb{C}$)

Symmetric:

Alternating: Symplectic geometry

3.1.2 Classification of bilinear forms in general

For any non-degenerate bilinear forms, there are two types: symmetric and alternating (skew-symmetric).

The study of symmetric bilinear forms yields to the standard geometry. ? It also leads to the theory of quadratic forms. A core question on this is to study how many non-degenerate bilinear forms over a field that are not isomorphic to each other. All such non-degenerate forms over a certain field forms a Witt ring. This topic occurs frequently in algebraic number theory and p -adic analysis.

The study of alternating bilinear forms is still leads to symplectic spaces.

3.1.3 A generalization of inner product

Inner product on a $\mathbb{R}$ -vector space or a $\mathbb{C}$ -vector space V is the most common one. Now, $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ ($F = \mathbb{R}$ or $\mathbb{C}$). It satisfies three properties:

- (1) $\langle u, v \rangle = \overline{\langle v, u \rangle}$
- (2) bilinear in each slot.
- (3) positive definite: $\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0 \Leftrightarrow u = 0$.

By virtue of inner product, the notions of length, angle and orthogonality are definable. These terms encompass 'geometry' world. But, in general, people want to do geometry not only on $\mathbb{R}$ and $\mathbb{C}$, but also on other fields, or even rings, like $\mathbb{Q}_p$, $\mathbb{Q}$. On general rings (even fields), notions of order and conjugate do not exist. This pushes us to generalize inner product in some sense. More precisely, the property (1) and (3) needs to be substituted.

For (1), what kind of properties does a conjugate map hold? Notice that $\bar{\cdot} : \mathbb{C} \rightarrow \mathbb{C}$ is an automorphism of ring $\mathbb{C}$ and $\bar{\bar{z}} = z$ for any $z \in \mathbb{C}$. Thereby, for a free R -module M , we can introduce an extra map called *involution*

Definition 3.1: Involution

An **involution** of a ring R is a homomorphism $\sigma : R \rightarrow R$ with $\sigma^2 = \text{id}_R$.

Remark • Consider a finite field $\mathbb{F}_{p^2}$. Then, the Frobenius map $\text{Frob}_p : \mathbb{F}_{p^2} \rightarrow \mathbb{F}_{p^2}$ is an involution.

For (3), since we have no order, one way to generalize it is to guarantee that the ruler f is not ineffective, that is f is not always zero. This is the **non-degeneracy**:

$$\forall u \in M - \{0\}, \exists v \in M, f(u, v) \neq 0$$

Thereby, a notion for inner product of a free module can be given as follows:

Definition 3.2: Inner product for a free R -module

Let M be a free R -module. An **inner product** on M is a *non-degenerate sesquilinear* map $f : M \times M \rightarrow R$ together with an involution map $\sigma : R \rightarrow R$ satisfying the σ -Hermitian form $\forall u, v \in M, f(u, v) = \sigma(f(v, u))$.

With involution, one can also mimic the Hermitian form to define an R -module version. Keeping the same settings, an **Hermitian form** is defined to be

3.2 Involution

3.2.1

There is a correspondence between the bilinear map $B : V \times V \rightarrow k$ and the isomorphism $\iota : V \rightarrow V^\vee$:

$$\begin{array}{ccc} B : V \times V \rightarrow k & \xrightarrow{\quad} & \iota : V \rightarrow V^\vee \\ & \swarrow \quad \searrow & \\ (v, w) \mapsto \iota(v)(w) & & u \mapsto B(u, -) \end{array}$$

To consider a bilinear map $B : V \times V \rightarrow k$, it suffices to think the linear map $\tilde{B} : V \otimes V \rightarrow k$. Then, there is a correspondence between the collection of all $\tilde{B}$ and all ι , namely

$$\text{Hom}_k \mathbf{Vect}(V \otimes V, k) \xleftarrow{\sim} \text{Hom}_k \mathbf{Vect}(V, V^\vee)$$

Notice that $V^\vee = \text{Hom}_k(V, k)$. This isomorphism between two Hom-sets is actually

$$\text{Hom}_k \mathbf{Vect}(V \otimes V, k) \xleftarrow{\sim} \text{Hom}_k \mathbf{Vect}(V, \text{Hom}_k(V, k))$$

which is a realization of Hom- $\otimes$ adjoint property.

When it comes to inner product, the reason why ι is superior to inner product is **why**

3.2.2 Orthogonality

The classic orthogonality is translated into the following. Given an inner product space $\langle \cdot, \cdot \rangle$, it has a corresponding $\iota : V \rightarrow V^\vee$. Given two vectors $u, v \in V$ that are orthogonal, $\iota(u)(v) = \langle u, v \rangle = 0$. So, $v \in \ker(\iota(u))$.

For a subspace $S \leq V$, the **orthogonal complement** $S^\perp$ of S , is correspondingly

$$S^\perp = \{v \in V : \forall u \in S, \langle u, v \rangle = 0\} = \bigcap_{u \in S} \ker(\iota(u))$$

3.3 Inner product space over $\mathbb{C}$

3.3.1 Spectral theorem

3.3.2 Reconstruction of operators

4 Exterior algebra

4.1 Basic notions

Consider a group action $S_n \curvearrowright V^n$ by $\sigma \in S_n, \forall \mathbf{v} := (v_1, \dots, v_n) \in V^n$,

$$\sigma \mathbf{v} := (v_{\sigma(1)}, \dots, v_{\sigma(n)})$$

This is a group action. For id, the action is trivial. For $\tau, \sigma \in S_n, \tau(\sigma \mathbf{v}) = \tau(v_{\sigma(1)}, \dots, v_{\sigma(n)}) = (v_{\tau\sigma(1)}, \dots, v_{\tau\sigma(n)})$. Now, let's consider another action S_n on the collection of all multilinear maps from V^n to W : $\{\psi : V^n \rightarrow W : \psi \text{ is a multilinear map}\}$, by taking arbitrary $\sigma \in S_n$ and ψ ,

$$\sigma \psi := \left(\mathbf{v} \mapsto \psi(\sigma \mathbf{v}) \right)$$

4.2 Symmetric multilinear map

4.3 Alternating multilinear map

The motivation for exterior algebra originates from Grassmann. He first tried to use a form to unify all the 'rigid' objects in various dimensional, for instance, lines, plane and paralleloids in $\mathbb{R}^3$. His idea was not valued until many years later when Cartan studied the **??**.

Given a vector space V over a field F , the intuition of an exterior algebra of power i (with $i \leq \dim_F V$), $\bigwedge^i(V)$, might be all the possible i -dimensional subspaces of V spanned by i elements in some basis of V . More concretely, for $\bigwedge^2(\mathbb{R}^3)$, it consists of all the elements that could span the xy , xz and yz planes.

To make the definition more comprehensive. Instead of a vector space, the context is put into free R -modules. For the heuristic reason, let's use the vector space to discuss and get better motivations.

Definition 4.1: Exterior Algebra

Let R be a commutative ring and M be a free R -module. The exterior algebra, denoted $\bigwedge(M)$, and a unital R -algebra homomorphism $i : M \rightarrow \bigwedge(M)$ are the unique ones satisfies: $\forall$ unital R -algebra A and $\forall$ R -linear map $\varphi : M \rightarrow A$ with $\forall v \in V, \varphi(v)\varphi(v) = 0$, $\exists!$ R -linear homomorphism $\Phi : \bigwedge(M) \rightarrow A$ such that $\Phi \circ i = \varphi$, i.e. the following diagram commutes:

$$\begin{array}{ccc} M & \xrightarrow{i} & \bigwedge(M) \\ & \searrow \varphi & \downarrow \exists! \Phi \\ & & A \end{array}$$

Construction of an exterior algebra: To construct such a most general R -algebra, start with the most general associative R -algebra containing M , the tensor algebra $T(M)$. Then, force the alternating property by taking a quotient. Take I to be the two-sided ideal of $T(M)$ generated by $\{m \otimes m : m \in M\}$. Then, the quotient

$$T(M)/I$$

is a candidate for the exterior algebra of M . ¹

4.4 Wedge product

4.5 Gradation of exterior algebras

4.5.1 k -th exterior power

For each gradation component $\bigwedge^k(M)$ of $\bigwedge(M)$, it can be depicted by another universal property. But notice that when we study $\bigwedge(M)$, it is always be considered as an R -algebra. But its pieces, $\bigwedge^k(M)$ loses the multiplication. Because elements $\alpha, \beta \in \bigwedge^k(M)$ are not closed under multiplication $\wedge$: $\alpha \wedge \beta \in \bigwedge^{2k}(M)$. So, usually $\bigwedge^k(M)$ is only considered as an R -module. This is also reflected in the universal property below.

Definition 4.2: Universal property of k -th exterior power

Let R be a commutative ring and M, N be free R -modules. The k -th **exterior power** of M , $\bigwedge^k(M)$, is defined to be the unique object with unique alternating multilinear $i : M^k \rightarrow \bigwedge^k(M)$ satisfying the following universal property: $\forall$ alternating multilinear map $\varphi : M^k \rightarrow N$, $\exists!$ R -module homomorphism $\Phi : \bigwedge^k(M) \rightarrow N$ such that $\Phi \circ i = \varphi$, i.e. the following diagram commutes:

$$\begin{array}{ccc} M^k & \xrightarrow{i} & \bigwedge^k(M) \\ & \searrow \varphi & \downarrow \exists! \Phi \\ & & N \end{array}$$

Proposition 4.1: Different spans of the ideal

$m \otimes m$ can be written in another form $m_1 \otimes \cdots \otimes m_n$ with $m_i = m_j$ for some $i \neq j$.

4.5.2 Basis of $\bigwedge^k(M)$

From now on, the center is to study the basic properties $\bigwedge^k(M)$ as an R -module.

¹The quotient is still an algebra, because $T(M)/I$ is a ring and I can be viewed as an R -submodule of M and $T(M)$, hence $T(M)/I$ has all structures.

Corollary 4.1: Basis of gradation components $\bigwedge^k(M)$

Let M be a free R -module over a commutative ring R with basis $\{m_1, \dots, m_n\}$. Then, one basis of $\bigwedge^k(M)$ is

$$\{m_{i_1} \wedge \dots \wedge m_{i_k} : 1 \leq i_1 < i_2 < \dots < i_k \leq n\}$$

When $k > n$, $\bigwedge^k(M) = \{0\}$. In particular, $\text{rank}_R \bigwedge^k(M) = \binom{n}{k}$.

Remark Especially, when $k = n$, $\bigwedge^n(M) = R(m_1 \wedge m_2 \wedge \dots \wedge m_n)$.

If M is not a free R -module, then it is probably false to have $\bigwedge^k(M) = 0$ for $k > n$. Here is an example. Let $R = \mathbb{Z}$ and consider $\mathbb{Z}[x, y]$. The ideal $\langle x, y \rangle$ is an R -submodule of $\mathbb{Z}[x, y]$, but it is not free because $x \cdot y - y \cdot x = 0$. We want to show that $I \wedge I \neq 0$.

First construct a map $\varphi : I \times I \rightarrow R/I \cong \mathbb{Z}$ by $(ax + by, cx + dy) \mapsto (ad - bc) + I$. This is a well-defined non-trivial alternating bilinear map.² By the universal property of k -component of $\bigwedge(M)$, $\exists!$ non-trivial R -linear map $\Phi : I \wedge I \rightarrow R/I$. But this implies that $I \wedge I \neq 0$.

Proof:

Example (1) Let V be a 1-dimensional vector space over F with basis element v . Then, every element of $\bigwedge^k(V)$ is $a_1 v \wedge a_2 v \wedge \dots \wedge a_k v = (a_1 a_2 \dots a_k) v \wedge \dots \wedge v$. Since $v \wedge v = 0$, $\bigwedge^0(V) = F$, $\bigwedge^1(V) = Fv = V$, $\bigwedge^i(V) = 0$ for $i \geq 2$. As a graded algebra, $\bigwedge(V) = F \oplus V \oplus 0 \oplus \dots$.

(2) Let V be a vector space with $\dim_F(V) = 2$, with basis element v and w . Then, every element is $(a_1 v + b_1 w) \wedge \dots \wedge (a_k v + b_k w)$. It follows that $\bigwedge^i(V) = 0$ for $i \geq 3$ because either v or w appear twice in every term of the expansion. For $\bigwedge^2(V)$, $\bigwedge^2(V) = T^2(V)/I$ where I is the ideal generated by the set $\{x \otimes x : x \in V\}$. $T^2(V)$ has a basis consisting of $v \otimes v$, $v \otimes w$, $w \otimes v$ and $w \otimes w$. But every element in I is

$$(av + bw) \otimes (av + bw) = a^2(v \otimes v) + ab(v \otimes w + w \otimes v) + b^2 w \otimes w$$

Notice that $v \otimes v$, $v \otimes w + w \otimes v$, $v \otimes w$, $w \otimes w$ is another basis for $T^2(V)$. As a subspace, I contains three elements of the basis of $T^2(V)$ but $v \otimes w \notin I$. So, it is 3-dim and $\bigwedge^2(V)$ is 1-dim. Further, $v \otimes w \notin T^2(V) \Rightarrow v \wedge w \neq 0 \in T^2(V)$, which gives $\bigwedge^2(V) = F(v \wedge w)$. As a graded algebra, $\bigwedge^2(V) = F \oplus V \oplus F(v \wedge w) \oplus 0 \oplus \dots$.

4.5.3 Characterizing by a vector space

Consider the space of multilinear alternating R -linear maps $M^k \rightarrow R$, denoted $\text{Alt}_R(M^k, R)$. By the universal property of k -the exterior algebras 4.2, there is a natural equivalence³

$$\text{Alt}_R(M^k, R) \cong \left(\bigwedge^k(M) \right)^*$$

As long as M is a finitely generated R -module, its dual module $M^* = \text{Hom}_R(M, R)$ is guaranteed to be a free-module and $M \cong M^*$. Under this condition,

$$\bigwedge^k(M^*) \cong \left(\bigwedge^k(M) \right)^*$$

4.5.4 Interior product

There is already an operation for lifting the degree of $\bigwedge^k(V)$. Another dual operation for degrading the operation is interior product. From now on, let $M^* := \text{Hom}_R(M, R)$ be the dual module of M , where R is a commutative ring. The goal is to find a mapping $\bigwedge^k(M) \rightarrow \bigwedge^{k-1}(M)$ and how this map interacts with the wedge product. Let's just observe the basic cases, $k = 0, 1, 2$, and then generalize this rule to arbitrary k .

²Well-defined: Suppose another representation of $(ax + by, cx + dy)$ is $(a'x + b'y, c'x + d'y)$, then $a' = a + yf$ and $b' = b - xf$ for some $f \in \mathbb{Z}[x, y]$. This is because $(a - a')x = (b' - b)y$ and it must be $x|b' - b$ and $y|a - a'$. Take $f = \frac{b-b'}{x}$.

Non-trivial: $\varphi(x, y) = 1 + I$ is not zero.

Bilinear and alternating follows easily from the definition of φ .

³The map $I : \text{Alt}_R(M^k, R) \cong \left(\bigwedge^k(M) \right)^*$ given by $f \mapsto \Phi_f$ is well-defined by the universal property applying to f , where $f : M^k \rightarrow R$ and Φ_f is the unique R -linear map corresponding to f . For two morphisms f and g , if $\Phi_f = \Phi_g$, they must agree on M^k : $f = \Phi_f \circ i = \Phi_g \circ i = g$. So, I is injective. Surjectivity comes from composing Φ with i . The linearity also comes from the R -linearity and uniqueness of Φ_f .

A natural way of degrading is to pick a $u \in M^*$, $v \in M$, and evaluate u on v : $u(v) \in R$. This map should analogue such a process. Keep $u \in M^*$ and name the map $i_u^k : \bigwedge^k(M) \rightarrow \bigwedge^{k-1}(M)$.

For $k = 0$ and $v \in \bigwedge^0(M)$, set $i_u^0(v) := 0$ always. For $k = 1$ and $\bigwedge^1(M)$, set $i_u^1(v) := u(v) \in R = \bigwedge^0(M)$.

In fact, there is a unique extension of $\bigwedge^1(M) \rightarrow \bigwedge^0(M)$ to $\bigwedge^k(M) \rightarrow \bigwedge^{k-1}(M)$. To avoid a degree k , the map i_u^k can also be regarded as a product, called *interior product*, denoted $u \lrcorner v$. We have

$$u \lrcorner v = i_u^k(v)$$

Let's continue extending the meaning of this $\lrcorner$. For $k = 2$, take $x, y \in \bigwedge^1(M)$. Then, we have $x \wedge y \in \bigwedge^2(M)$. $\lrcorner$ should be the unique product that extending $\lrcorner : M^* \times \bigwedge^1(M) \rightarrow \bigwedge^0(M)$. This is equivalent to seeking the unique product $\lrcorner : M^* \times \bigwedge^k(M) \rightarrow \bigwedge^{k-1}(M)$ such that: (1) $u \lrcorner v = u(v)$ for any $v \in \bigwedge^1(M)$ and (2) $\lrcorner(u, \bigwedge^k(M)) \subseteq \bigwedge^{k-1}(M)$.

For $u \lrcorner (x \wedge y)$, it is a vector in $\bigwedge^1(M)$ and dependent of x and y . To make i_u^2 linear, $\lrcorner(x \wedge y)$ should be a linear combination of x and y ,

$$u \lrcorner (x \wedge y) = c_1 x + c_2 y$$

where c_1 and c_2 also depend on x and y . Applying i_u^1 , $i_u^1 \circ i_u^2(x \wedge y) = c_1 u(x) + c_2 u(y) = 0$. One plausible choice of coefficients is $c_1 = -u(y)$ and $c_2 = u(x)$. So,

$$u \lrcorner (x \wedge y) = -u(y)x + u(x)y = u(x) \wedge y - u(y) \wedge x = (u \lrcorner x) \wedge y + (-1)x \wedge (u \lrcorner y)$$

- The first generalization admitted is the interaction between $u \in M^*$ and $x \wedge y$ where $x \in \bigwedge^r(M)$ and $y \in \bigwedge^s(M)$,

$$u \lrcorner (x \wedge y) = (u \lrcorner x) \wedge y + (-1)^r x \wedge (u \lrcorner y)$$

- The second generalization is for $u \in \bigwedge^l(M^*)$.

An easier and more acceptable way to establish the interior product is use the identification $\text{Alt}_R(M^k, R) \cong (\bigwedge^k(M))^*$.

Theorem 4.1: Axiomatic characterization and properties of interior product

Let M be a finitely generated free R -module. The interior product on $\bigwedge^k(M)$ satisfies the following properties $\forall u \in M^*$ and $\forall v \in M$

- (1) $\forall k, \iota_u : \bigwedge^k(M) \rightarrow \bigwedge^{k-1}(M)$
- (2) $\iota_u(v) = u(v)$
- (3) It satisfies the graded derivation of degree -1 : $\forall x \in \bigwedge^r(M)$ and $\forall y \in \bigwedge^s(M)$,

$$\iota_u(x \wedge y) = \iota_u(x) \wedge y + (-1)^{\deg x} x \wedge \iota_u(y)$$

4.5.5 Application to algebraic geometry: Plücker embeddings

Given a vector space V over F , with $\dim_F(V) = n$. The projective space $\mathbb{P}(V)$ parametrises all the lines of V (all one-dimensional subspace of V). In more condensed language,

$$\mathbb{P}(V) \rightsquigarrow \{L_1 : L_1 \leq V, \dim_F L_1 = 1\}$$

The collection on the right hand side is exactly the Grassmannian or Grassmann variety $\text{Gr}_1(V)$. Now the goal is to generalize this ideal to any $\text{Gr}_r(V)$ for $r \leq n$. Are there any counterparts(projective spaces) corresponding to $\text{Gr}_r(V)$?

Since all the r -dimensional subspaces are taken into consideration, the exterior product $\bigwedge^r(V)$ is a huge vector space that contains every possible r -dimensional subspaces. There should be a map $\text{Gr}_r(V) \rightarrow \mathbb{P}(\bigwedge^r(V))$. But as in the case of V and $\mathbb{P}(V)$, V itself has many elements that describe the same information(the same line) that are compressed into one in $\mathbb{P}(V)$. To remove the redundancy in $\bigwedge^r(V)$, its projective space is better than itself. The effective map should be

$$\text{Gr}_r(V) \rightarrow \mathbb{P}\left(\bigwedge^r(V)\right)$$

To define this map concretely, start from an r -dimensional subspace L_r of V , pick a basis $\{e_1, \dots, e_r\}$ for this L_r and send this basis to $e_1 \wedge \dots \wedge e_r$. Finally, take the equivalent class $[e_1 \wedge \dots \wedge e_r] \in \mathbb{P}(\bigwedge^r(V))$. So, this map is defined elementwise as

$$L_r = \text{span}\{e_1, \dots, e_r\} \mapsto P_{L_r} := [e_1 \wedge \dots \wedge e_r]$$

• This map is well-defined or independent of the choice of basis. Take another basis of L_r , $\{f_1, \dots, f_r\}$. This map gives another representation $[f_1 \wedge \dots \wedge f_r]$. Both $e_1 \wedge \dots \wedge e_r$ and $f_1 \wedge \dots \wedge f_r$ are elements of $\bigwedge^r(L_r)$. Consider a base-change map $T : L_r \rightarrow L_r$ by $e_i \mapsto f_i$ for all i . By functoriality of $\bigwedge^r$, this T induces a linear map $\bigwedge^r(T) : \bigwedge^r(L_r) \rightarrow \bigwedge^r(L_r)$ such that

$$\bigwedge^r(T)(e_1 \wedge \dots \wedge e_r) := T(e_1) \wedge \dots \wedge T(e_r) = f_1 \wedge \dots \wedge f_r$$

In the meanwhile, $\bigwedge^r(T)(e_1 \wedge \dots \wedge e_r) = \det(T)e_1 \wedge \dots \wedge e_r$. Hence, $[f_1 \wedge \dots \wedge f_r] = [\det(T)e_1 \wedge \dots \wedge e_r]$.

4.6 Functoriality of exterior algebras

4.6.1 Morphisms between two exterior algebras

Let M and N be two free modules over a commutative ring R and $f : M \rightarrow N$ be a R -linear map. Apply the universal property in 4.1 to M and N .

$$\begin{array}{ccc} M & \xrightarrow{i} & \bigwedge(M) \\ & \searrow \varphi & \downarrow \exists! \Phi \\ & & \bigwedge(N) \end{array}$$

The unique R -algebra homomorphism Φ can be assigned a new symbol $\bigwedge(f)$.

5 Duality of inner product and exterior product